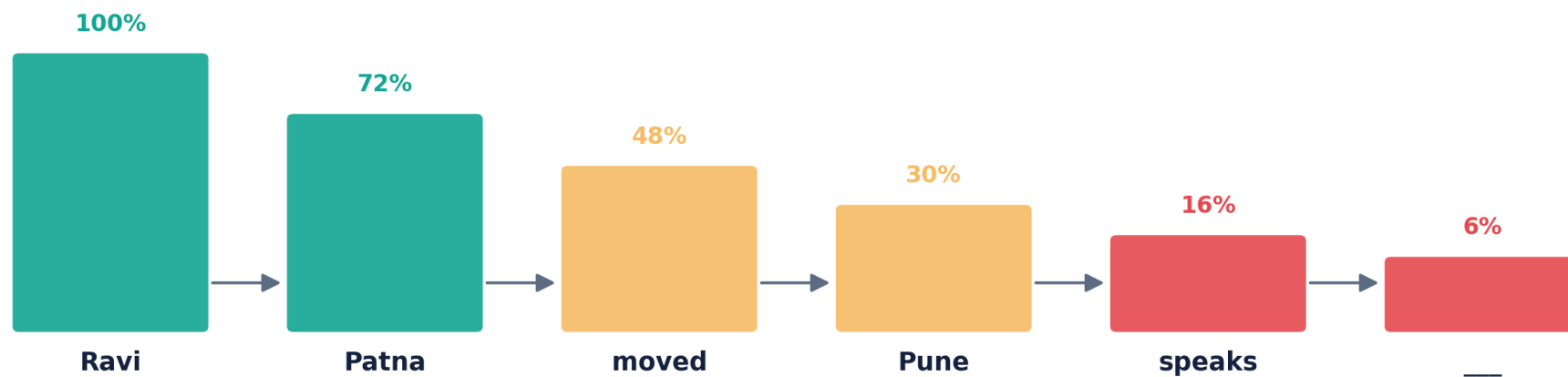


# Memory that fades

## MEMORY THAT LEAKS

"Ravi, who grew up in Patna and later moved to Pune, speaks \_\_\_\_"

how much of each word still survives in memory



**By the blank, "Patna" is nearly gone — so it cannot tell you Hindi**

**That fading is the vanishing gradient. Attention was invented to fix it.**

*Early words fade as the sentence grows*

An RNN reads a sentence word by word, carrying a **memory** forward. That memory leaks.

- 1 Each word updates a small hidden state passed to the next step.
- 2 By word fifty, the influence of word one has almost vanished.
- 3 That fading is the vanishing gradient — the reason attention was invented.

#### REMEMBER

**RNNs taught us sequence matters. Transformers fixed the forgetting.**

